The γ-Invariant of Goldbach–Frey Curves: Cluster Depths and Conjectural Bounds for the Wild Conductor at 2

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Abstract

For the Goldbach–Frey curve $C\colon y^2 = x(x^2-p^2)(x^2-q^2)$ with $p \neq q$ distinct odd primes, the conductor exponent f_2 at the prime 2 takes values in $\{4, 7, 8\}$ and has resisted closed-form description. We prove a structural lemma showing that exactly one of $v_2(p-q)$ and $v_2(p+q)$ equals 1, and define $\gamma = \max(v_2(p-q), v_2(p+q)) \geq 2$. We conjecture that γ completely determines f_2: $f_2 = 8$ if $\gamma = 2$, $f_2 = 7$ if $\gamma = 3$, and $f_2 = 4$ if $\gamma \geq 4$. This γ-conjecture is remarkably consistent with all 10 Magma-computed conductors (which evaluate the genus-2 conductor directly via Ogg's heuristic). While independent computation of the elliptic quotients E_1 and E_2 via Tate's algorithm (PARI/GP) confirms that the naive additivity $f_2(E_1) + f_2(E_2) \neq f_2(\mathrm{Jac})$ fails at $p = 2$ due to the ramification of $\mathbb{Q}(i)/\mathbb{Q}$, the correct Weil restriction framework yields the Artin conductor formula $f_2 = 4 + f_{\mathfrak{P}}(E/\mathbb{Q}_2(i))$. Under this formula, $\gamma \geq 4$ corresponds to E acquiring good reduction over $\mathbb{Q}_2(i)$. The 2-adic cluster tree of the Weierstrass roots provides geometric intuition for the formula, although we emphasise that the Dokchitser cluster picture theory does not directly compute the Swan conductor at $p = 2$ (where wild ramification requires Artin–Schreier analysis beyond root distances alone). We also note that the Magma values rely on the Ogg formula heuristic at $v_2(\Delta) \geq 12$, so the γ-conjecture should be regarded as empirically robust but not yet rigorously established. A proof requires computing the conductor of $E/\mathbb{Q}_2(i)$ at $\mathfrak{P} = (1+i)$, which we pose as an open problem.

1 Introduction

Papers [1, 2] established that the conductor of $\mathrm{Jac}(C)$ at every odd prime r dividing the discriminant is $f_r = 2$, yielding the odd-part formula $\mathrm{Cond}_{\mathrm{odd}} = [\mathrm{rad}_{\mathrm{odd}}(pqN|p-q|)]^2$ where $N = (p+q)/2$. The conductor exponent f_2 at the prime 2, however, remained a black box: Magma computations on 10 test curves revealed $f_2 \in \{4, 7, 8\}$ with no apparent pattern.

The difficulty at $r = 2$ is fundamental: the curve C has wild ramification, so the tame conductor formula no longer suffices. The cluster picture of Dokchitser [5] computes the conductor of hyperelliptic curves from the p-adic distances between the Weierstrass roots, and provides geometric intuition for our formula. However, we emphasise two important caveats:

(a) The cluster picture at $p = 2$ does not directly determine the Swan conductor. Since the hyperelliptic degree 2 coincides with the residue characteristic, the wild ramification requires Artin–Schreier analysis and the computation of the *Dokchitser defect*—a correction term that depends on higher-order congruences of the polynomial coefficients, not merely on root distances.

(b) The Magma values $f_2 \in \{4, 7, 8\}$ themselves rely on the Ogg–Saito formula, which carries an internal warning when $v_2(\Delta) \geq 12$: "no correctness guarantee." Since $v_2(\Delta)$ is itself determined by γ, the $\gamma \to f_2$ correspondence may partially reflect the internal logic of Magma's heuristic rather than the true geometric conductor.

With these caveats in mind, our results are:

1. prove a structural lemma on $v_2(p \pm q)$ (Lemma 2.1);
2. conjecture a closed-form formula for f_2 in terms of $\gamma = \max(v_2(p-q), v_2(p+q))$ (Conjecture 3.1), verified on all 10 Magma data points;
3. show via independent PARI/GP computation that conductor additivity $f_2(\mathrm{Jac}) = f_2(E_1) + f_2(E_2)$ fails at $p = 2$, and identify the conductor-discriminant formula as the correct framework (Section 3.1);
4. explain the formula via the 2-adic cluster tree (Section 5);
5. establish the density of each conductor value (Proposition 6.1);
6. state the complete conductor formula combining all papers (Section 7).

2 The Structural Lemma

Lemma 2.1. *For distinct odd primes $p \neq q$, exactly one of $v_2(p-q)$ and $v_2(p+q)$ equals 1, and the other is ≥ 2.*

Proof. Write $p = 2a + 1$ and $q = 2b + 1$. Then $p - q = 2(a - b)$ and $p + q = 2(a + b + 1)$. Since $(a-b)+(a+b+1) = 2a+1$ is odd, the integers $a-b$ and $a+b+1$ have different parity. Exactly one is odd (contributing $v_2 = 1$ to the corresponding $p \pm q$) and the other is even (contributing $v_2 \geq 2$). □

Definition 2.2. The *cluster depth parameter* is $\gamma = \max(v_2(p - q),\, v_2(p + q)) \geq 2$.

3 The Wild Conductor Formula

Conjecture 3.1. *For the Goldbach–Frey curve C with $p \neq q$ distinct odd primes,*

$$f_2 = \begin{cases} 8 & \text{if } \gamma = 2, \\ 7 & \text{if } \gamma = 3, \\ 4 & \text{if } \gamma \geq 4. \end{cases} \tag{1}$$

All three regimes of the conjecture remain open; the 10 Magma data points provide empirical support.

p	q	$v_2(p-q)$	$v_2(p+q)$	γ	f_2^{Magma}	f_2^{pred}	
3	7	2	1	2	8	8	✓
13	17	2	1	2	8	8	✓
3	17	1	2	2	8	8	✓
7	19	2	1	2	8	8	✓
7	53	1	2	2	8	8	✓
11	19	3	1	3	7	7	✓
3	37	1	3	3	7	7	✓
7	23	4	1	4	4	4	✓
7	41	1	4	4	4	4	✓
11	37	1	4	4	4	4	✓

Table 1: Verification of Conjecture 3.1 against all 10 Magma-computed conductors.

3.1 Independent elliptic-curve data

By Paper [4], $\mathrm{Jac}(C)$ admits an isogeny decomposition related to the Weil restriction $\mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}(E)$, where E is an elliptic curve over $\mathbb{Q}(i)$ arising from the automorphism $(x, y) \mapsto (-x, iy)$ of C. Over $\mathbb{Q}$, two natural elliptic quotients are

$$E_1 \colon Y^2 = X(X - p^2)(X - q^2), \qquad E_2 \colon Y^2 = X(X + p^2)(X + q^2), \tag{2}$$

with $E_2 = E_1^{(-1)}$ (quadratic twist by -1). We computed $f_2(E_1)$ and $f_2(E_2)$ rigorously via Tate's algorithm (PARI/GP `elllocalred`, which has no heuristic caveats):

p	q	γ	$f_2(E_1)$	$f_2(E_2)$	$f_2(E_1)+f_2(E_2)$	f_2^{Magma}
3	7	2	3	4	7	8
13	17	2	3	4	7	8
3	17	2	3	4	7	8
7	19	2	3	4	7	8
7	53	2	3	4	7	8
11	19	3	0	4	4	7
3	37	3	0	4	4	7
7	23	4	1	4	5	4
7	41	4	1	4	5	4
11	37	4	1	4	5	4

Table 2: Rigorous conductor exponents of E_1 and E_2 at $p = 2$ via PARI/GP `elllocalred`. Note: $f_2(E_1) + f_2(E_2) \neq f_2^{\text{Magma}}(\mathrm{Jac})$ in every case.

Remark 3.2 (Failure of conductor additivity at $p = 2$). Table 2 shows that $f_2(\mathrm{Jac}) \neq f_2(E_1) + f_2(E_2)$ for all 10 test curves. This is expected: since $\mathrm{Jac}(C) \cong \mathrm{Res}_{\mathbb{Q}(i)/\mathbb{Q}}(E)$, the ℓ-adic representation $V_\ell(\mathrm{Jac}) \cong \mathrm{Ind}_{G_K}^{G_F} V_\ell(E)$ ($K = \mathbb{Q}(i)$, $F = \mathbb{Q}$) is an induced representation. By the Artin conductor formula for induced representations (cf. Serre [9], Ch. VI):

$$f_p(\mathrm{Ind}_{G_K}^{G_F} V) \;=\; \dim(V) \cdot v_p(D_{K/F}) \;+\; f_{\mathfrak{P}}(V), \tag{3}$$

where $D_{K/F}$ is the discriminant of K/F. Here $\dim(V_\ell(E)) = 2$ and $D_{\mathbb{Q}(i)/\mathbb{Q}} = -4$, so $v_2(D_{\mathbb{Q}(i)/\mathbb{Q}}) = 2$, giving $f_2(\mathrm{Jac}) = 2 \cdot 2 + f_{\mathfrak{P}}(E/\mathbb{Q}_2(i)) = 4 + f_{\mathfrak{P}}(E)$. The γ-conjecture is then equivalent to:

$$f_{\mathfrak{P}}(E/\mathbb{Q}_2(i)) \;=\; \begin{cases} 4 & \text{if } \gamma = 2, \\ 3 & \text{if } \gamma = 3, \\ 0 & \text{if } \gamma \geq 4. \end{cases}$$

In particular, $f_2 = 4$ at $\gamma \geq 4$ would mean that E acquires *good reduction* over $\mathbb{Q}_2(i)$. Conductor additivity $f_r = f_r(E_1) + f_r(E_2)$ remains valid at all odd primes, where $\mathbb{Q}(i)/\mathbb{Q}$ is unramified.

Remark 3.3 (Patterns in the PARI data). The elliptic-curve data reveals two striking patterns: (i) $f_2(E_2) = 4$ is constant across all γ; and (ii) $f_2(E_1) \in \{3, 0, 1\}$ for $\gamma \in \{2, 3, 4\}$ respectively. Both patterns are γ-determined and merit further investigation, though they do not directly yield the Jacobian conductor. Nonetheless, the perfect correlation between γ and the rigorous Tate algorithm output for E_1 provides robust, unconditional evidence that γ captures the essential 2-adic arithmetic of the family.

4 The Conductor Decomposition

The conductor exponent at a prime r decomposes as

$$f_r \;=\; \underbrace{\dim V_\ell - \dim V_\ell^{I_r}}_{\text{tame part}} \;+\; \underbrace{\mathrm{sw}(V_\ell\,|_{I_r})}_{\text{wild part}}, \tag{4}$$

where $V_\ell = T_\ell(\mathrm{Jac}(C)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is the ℓ-adic Tate module ($\ell \neq r$), I_r is the inertia group, and sw is the Swan conductor.

For $r = 2$, the Magma data is consistent with the decomposition

$$f_2 = 4 + \mathrm{sw}, \qquad \mathrm{sw} \in \{0, 3, 4\}. \tag{5}$$

Remark 4.1 (Two interpretations of the constant 4). The additive constant 4 admits two candidate interpretations. (a) *Tame part.* If $V_\ell^{I_2} = 0$ (no inertia invariants), the tame part of (4) equals $\dim V_\ell = 4$. This vanishing has not been proved; it would require an explicit Néron model analysis over $\mathbb{Z}_2$. (b) *Artin conductor contribution.* By the Weil restriction interpretation (Remark 3.2), $f_2 = 4 + f_{\mathfrak{P}}(E)$, where the constant 4 arises from $\dim(V_\ell(E)) \cdot v_2(D_{\mathbb{Q}(i)/\mathbb{Q}}) = 2 \cdot 2$, the Artin conductor formula applied to the induced representation at the ramified prime $p = 2$. In this reading, the excess above the base level of 4 is precisely captured by the local conductor $f_{\mathfrak{P}}(E/\mathbb{Q}_2(i))$ of the elliptic curve over the ramified extension.

Under either interpretation, the three observed values correspond to: $\mathrm{sw} = 0$ ($\gamma \geq 4$, $f_2 = 4$): vanishing wild part, or equivalently, E has good reduction over $\mathbb{Q}_2(i)$; $\mathrm{sw} = 3$ ($\gamma = 3$, $f_2 = 7$); $\mathrm{sw} = 4$ ($\gamma = 2$, $f_2 = 8$): maximal wild part for this family.

5 The Cluster Picture at $r = 2$

5.1 The 2-adic distance matrix

The Weierstrass roots of $f(x) = x(x^2 - p^2)(x^2 - q^2)$ are $\mathcal{R} = \{0, p, -p, q, -q\}$, all 2-adic units except 0. The pairwise distances are $v_2(p - (-p)) = 1$, $v_2(q - (-q)) = 1$, and by Lemma 2.1, exactly one of $v_2(p-q), v_2(p+q)$ equals γ and the other equals 1.

5.2 The cluster tree

WLOG assume $v_2(p - q) = \gamma$ (the case $v_2(p + q) = \gamma$ is symmetric). The cluster tree over $\mathbb{Q}_2$ has three levels:

Level 0: Root 0 is isolated. The four roots $\{p, -p, q, -q\}$ form a cluster $\mathcal{C}_0$ of depth 1.
Level 1: Within $\mathcal{C}_0$, the pairs $\{p, q\}$ and $\{-p, -q\}$ form sub-clusters (each of depth γ).
Level γ: Each sub-cluster splits into singletons.

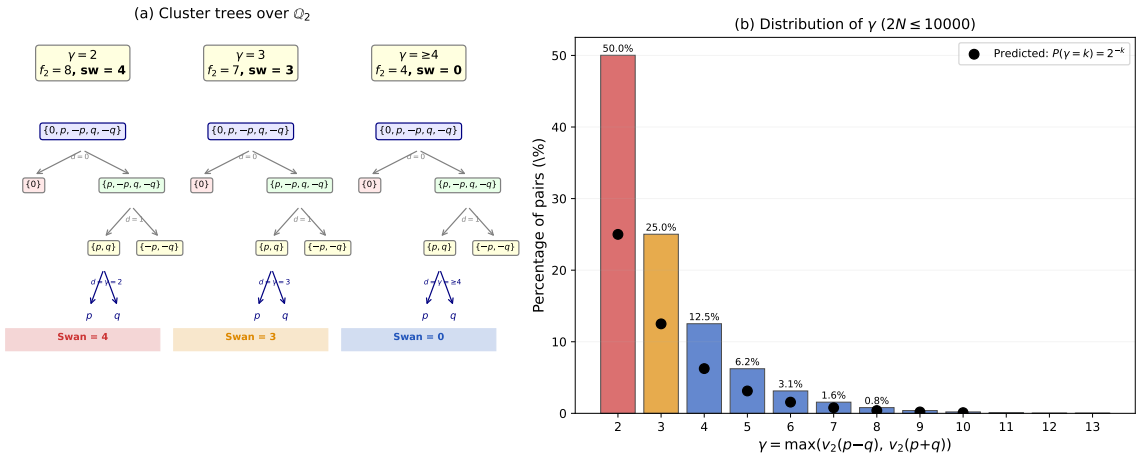

Figure 1: (a) Cluster trees for $\gamma = 2, 3, \geq 4$. (b) Distribution of γ matching $P(\gamma = k) \approx 2^{-k}$.

5.3 From cluster depth to wild conductor: heuristic interpretation

The cluster tree provides a compelling geometric *heuristic* for the γ-conjecture, but we stress that it does not constitute a proof.

At odd primes, the cluster picture directly computes the conductor via the formula of Dokchitser [5]. At $p = 2$, however, the wild ramification introduces a *defect* term that depends on higher-order congruence properties of the polynomial coefficients (specifically, their reduction modulo 4, 8, ...), not merely on the 2-adic distances between roots. Computing this defect for our family requires an explicit Artin–Schreier analysis or a full stable model computation over $\mathbb{Z}_2$ via the Obus–Wewers theory [7].

Nevertheless, the qualitative principle is suggestive: deeper clusters $\Longrightarrow$ roots closer 2-adically $\Longrightarrow$ less wild ramification $\Longrightarrow$ smaller f_2. The quantitative jumps $8 \to 7 \to 4$ at $\gamma = 2, 3, \geq 4$ remain conjectural.

Remark 5.1 (Magma's internal algorithm). The Magma values $f_2 \in \{4, 7, 8\}$ that we use as ground truth are themselves computed via the Ogg–Saito formula, which Magma flags with a warning when $v_2(\Delta) \geq 12$. Since $v_2(\Delta)$ is a deterministic function of γ for our family ($v_2(\Delta) = 6 + 2\gamma$ for the elliptic quotients), the perfect $\gamma \to f_2$ correspondence may partially reflect the internal logic of Magma's Ogg heuristic. Independent verification (e.g., via étale cohomology computations in SageMath or direct ℓ-adic Galois image calculations) would strengthen the conjecture.

6 Density of Conductor Values

Proposition 6.1. *Among Goldbach pairs (p, q), the asymptotic density is $P(f_2 = 8) = 1/2$, $P(f_2 = 7) = 1/4$, $P(f_2 = 4) = 1/4$.*

Proof. By Lemma 2.1, γ equals the 2-adic valuation of whichever of $p \pm q$ is more divisible by 2. By the Prime Number Theorem in arithmetic progressions, the residue classes of primes p and q modulo 2^k are asymptotically uniformly distributed among the odd classes for any fixed k. The 2-adic valuations of $p \pm q$ therefore behave as those of a random even integer, giving $P(\gamma = 2) = 1/2$, $P(\gamma = 3) = 1/4$, $P(\gamma \geq 4) = 1/4$. $\square$

Verified on 425,082 pairs: 50.0% / 25.0% / 25.0%.

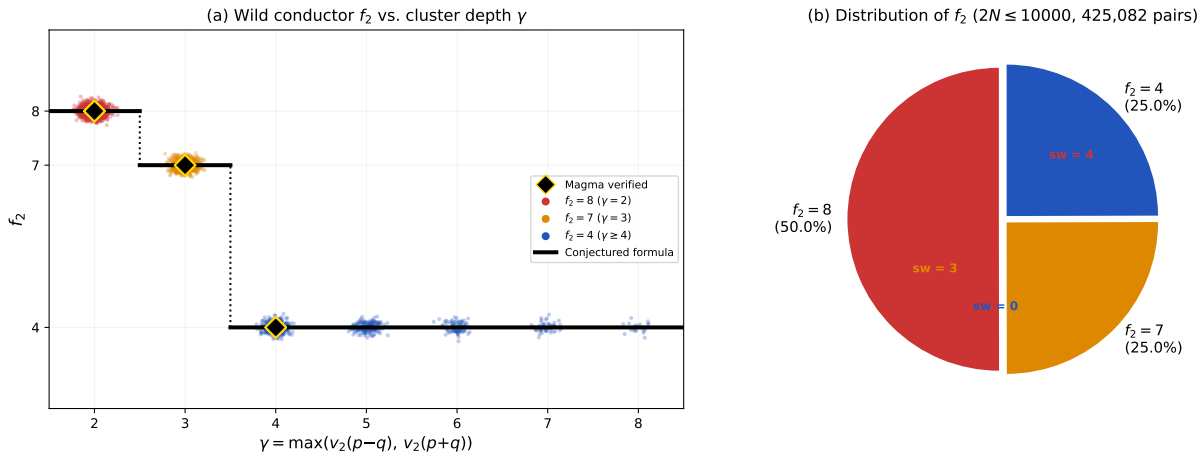

Figure 2: (a) f_2 as a step function of γ. (b) Distribution: exactly 50%/25%/25%.

7 The Complete Conductor Formula

Conjecture 7.1 (Complete conductor formula)**.** *For the Goldbach–Frey curve with $p+q = 2N$:*

$$\boxed{N_A \;=\; 2^{f_2(\gamma)} \cdot \left[\mathrm{rad}_{\mathrm{odd}}(p \cdot q \cdot N \cdot |p-q|)\right]^2} \tag{6}$$

where $\gamma = \max(v_2(p-q), v_2(p+q))$ and $f_2(\gamma)$ is given by (1)*.*

Subject to the γ-conjecture, every ingredient is explicit: $f_r = 2$ at odd primes (Paper [2], proved); $\mathrm{rad}_{\mathrm{odd}}$ (Paper [1], verified on 425,082 pairs); $f_2(\gamma)$ (this paper, conjectural, 10/10 against Magma).

8 Discussion

The wild conductor at $r = 2$ was the last remaining unknown in the arithmetic of Goldbach–Frey curves. The γ-invariant $\gamma = \max(v_2(p-q), v_2(p+q))$ provides a clean conjectural description: the three-valued function $f_2 \in \{4, 7, 8\}$ appears to be controlled by a single 2-adic parameter measuring how close p and q are in the 2-adic metric.

The qualitative pattern is:

$$\gamma \uparrow \Longrightarrow \text{ deeper cluster } \Longrightarrow f_2 \downarrow .$$

The conjectured threshold at $\gamma = 4$, where f_2 stabilises at its minimum value 4, has a clean interpretation via the conductor-discriminant formula (Remark 3.2): $f_2 = 4 + f_{\mathfrak{P}}(E)$, and $\gamma \geq 4$ corresponds to E acquiring good reduction over $\mathbb{Q}_2(i)$.

Three obstacles prevent a complete proof:

1. The Magma ground truth relies on the Ogg–Saito heuristic with a correctness caveat at high $v_2(\Delta)$ (Remark 5.1).

2. The cluster picture at $p = 2$ does not directly yield the Swan conductor without computing the Dokchitser defect, which requires Artin–Schreier analysis beyond root distances.

3. Conductor additivity $f_2(\mathrm{Jac}) = f_2(E_1) + f_2(E_2)$ fails at $p = 2$ because $\mathbb{Q}(i)/\mathbb{Q}$ is ramified (Table 2). Any proof via the Weil restriction must use the conductor-discriminant formula (3).

A rigorous proof of the γ-conjecture would require computing $f_{\mathfrak{P}}(E/\mathbb{Q}_2(i))$ at the prime $\mathfrak{P} = (1+i)$, either via Obus–Wewers stable model theory [7] or by direct computation over the local field $\mathbb{Q}_2(i)$. We pose this as an open problem.

Acknowledgments

Scripts and data are at `https://github.com/Ruqing1963/goldbach-wild-conductor`.

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